10-6 Q14 16 22
10-7 Q11 12

C10S06Q14

$H_0: \sigma_1^2 = \sigma_2^2$
$H_a: \sigma_2^2 > \sigma_1^2$

$F = \frac{28.2^2}{15.6^2} = 3.268$

p-value $= P(F_{29,29} \geq 3.268) = 0.001$

$0.001 < 0.05 \Rightarrow$ Reject H_0

C10S06Q16

$H_0: \sigma_1^2 = \sigma_2^2$
$H_a: \sigma_1^2 \neq \sigma_2^2$
$\alpha = 0.05$

$F_{stat} = \frac{114^2}{103^2} = 1.225$

p-value $= 2P(F_{14,14} > 1.225)$

F is large > 0.05
$\Rightarrow$ Non-reject H_0

C10S06Q22

(a) $H_0: \sigma_1^2 = \sigma_2^2$
$H_a: \sigma_1^2 \neq \sigma_2^2$
$\alpha = 0.05$

$F = \frac{2.96}{1.54} = 1.922$

F is large > 0.05
$\Rightarrow$ Non-reject H_0

p-value $= 2P(F_{14,14} \geq 1.922)$

(b) $\alpha = 0.01$

$$\frac{s_1^2}{s_2^2} \cdot \frac{1}{F_{14,14,0.005}} \leq \frac{\sigma_1^2}{\sigma_2^2} \leq \frac{s_1^2}{s_2^2} \cdot F_{14,14,0.05}$$

$= (0.128, 2.118)$

C10S07Q11

$\bar{x}_1 = 1.875$ $\bar{x}_2 = 2.625$
$s_1 = 0.8345$ $s_2 = 0.9161$
$\alpha = 0.05$

d	-3	1	-1	-2	1	-1	0	-1

$\bar{d} = -0.75$
$s_d = 1.3887$

$H_0: \mu_1 - \mu_2 = \mu_d = 0$
$H_a: \mu_1 - \mu_2 \neq 0$

$$t = \frac{\bar{d} - (\mu_1 - \mu_2)}{\frac{s_d}{\sqrt{n}}} = \frac{-0.75}{1.3887/\sqrt{8}}$$

$= -1.53$

$t_{7, 0.025} = 2.365$

$|-1.53| < 2.365 \Rightarrow$ Non-reject H_0

C10S07Q12

d	
d_1	-1
d_2	-1
	-2
	1
	-1
	-1
	0
	1

$\bar{d} = -0.75$
$s_d = 0.8864$

$H_0: \mu_d = 0$
$H_a: \mu_d \neq 0$
$\alpha = 0.05$

$$t = \frac{\bar{d}}{s_d/\sqrt{n}} = \frac{-0.75}{0.8864/\sqrt{8}}$$

$= -2.3931$

$t_{7, 0.025} = 2.365$

$|-2.3931| > 2.365 \Rightarrow$ Reject H_0